

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug



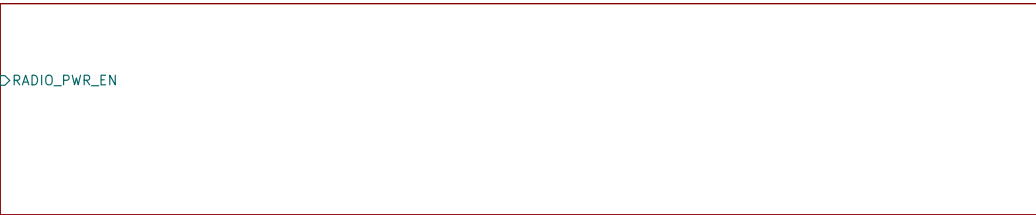
File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

ESP32-C6 radio

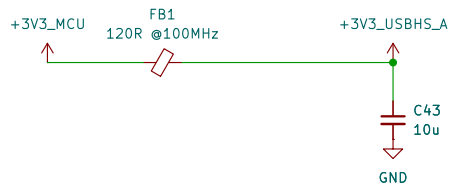


File: radio_esp32.kicad_sch

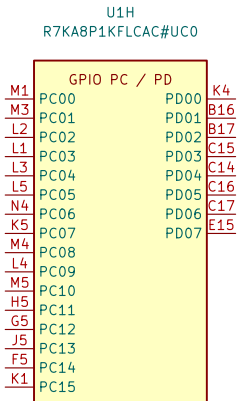
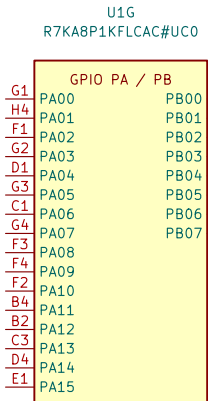
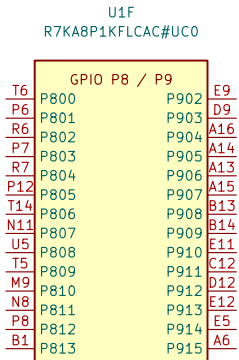
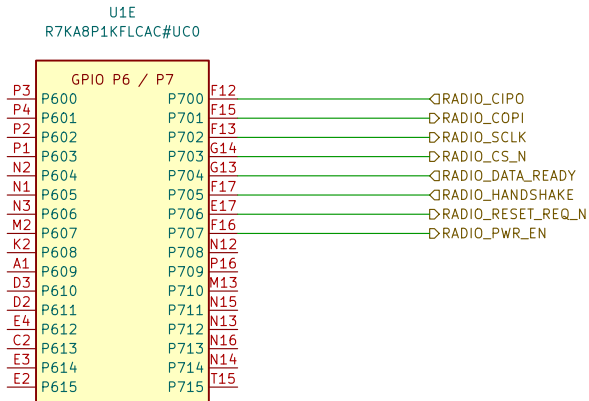
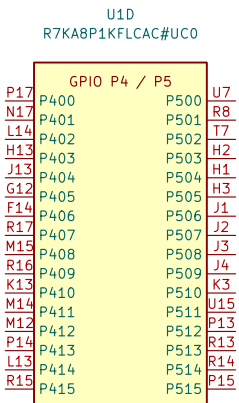
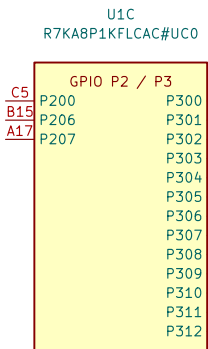
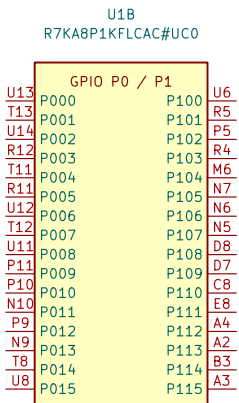
RADIO_PWR_EN

Sheet: /		
File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader - schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5		Id: 1/6

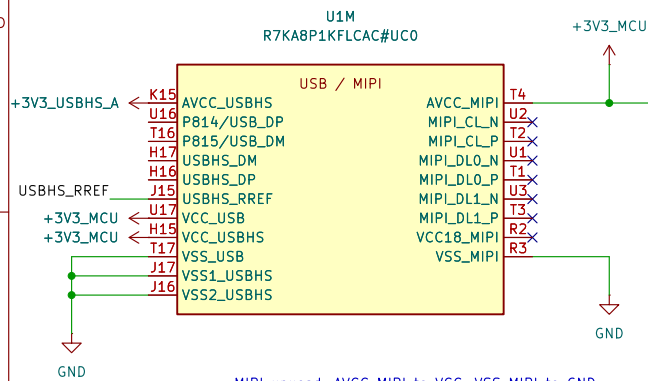
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



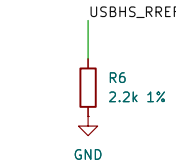
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



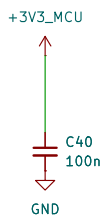
RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



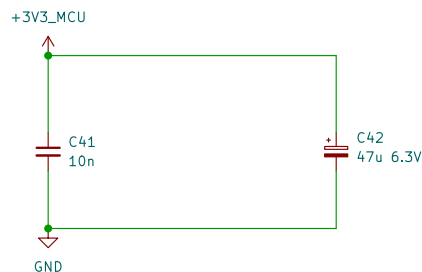
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1.H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

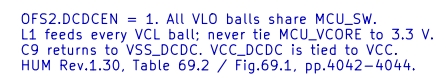
Size: A3

Date: 2026-09-05

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

Id: 2/6



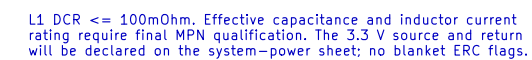
U1K
R7KA8P1KFLCAC#U00

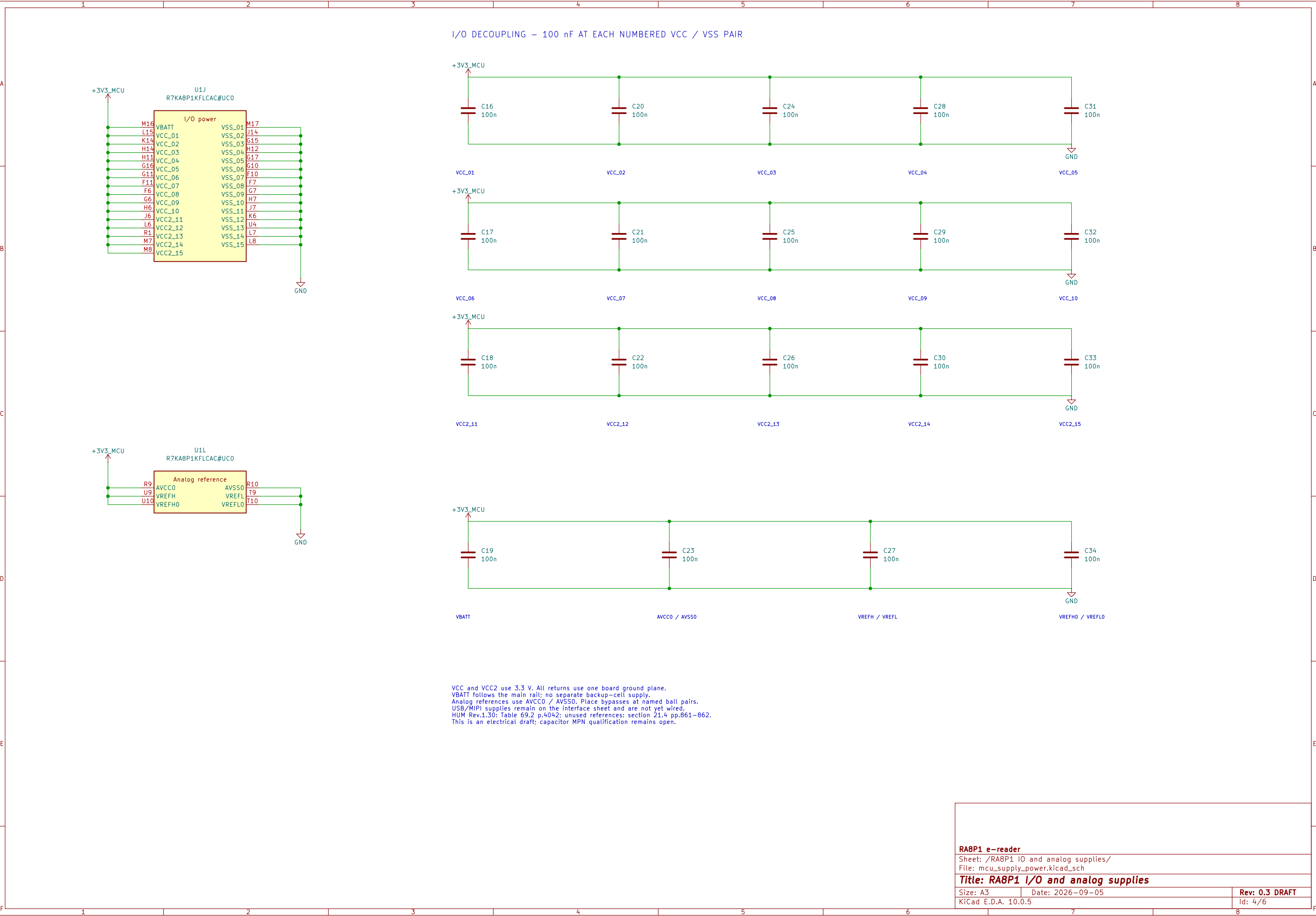
+3V3_MCU

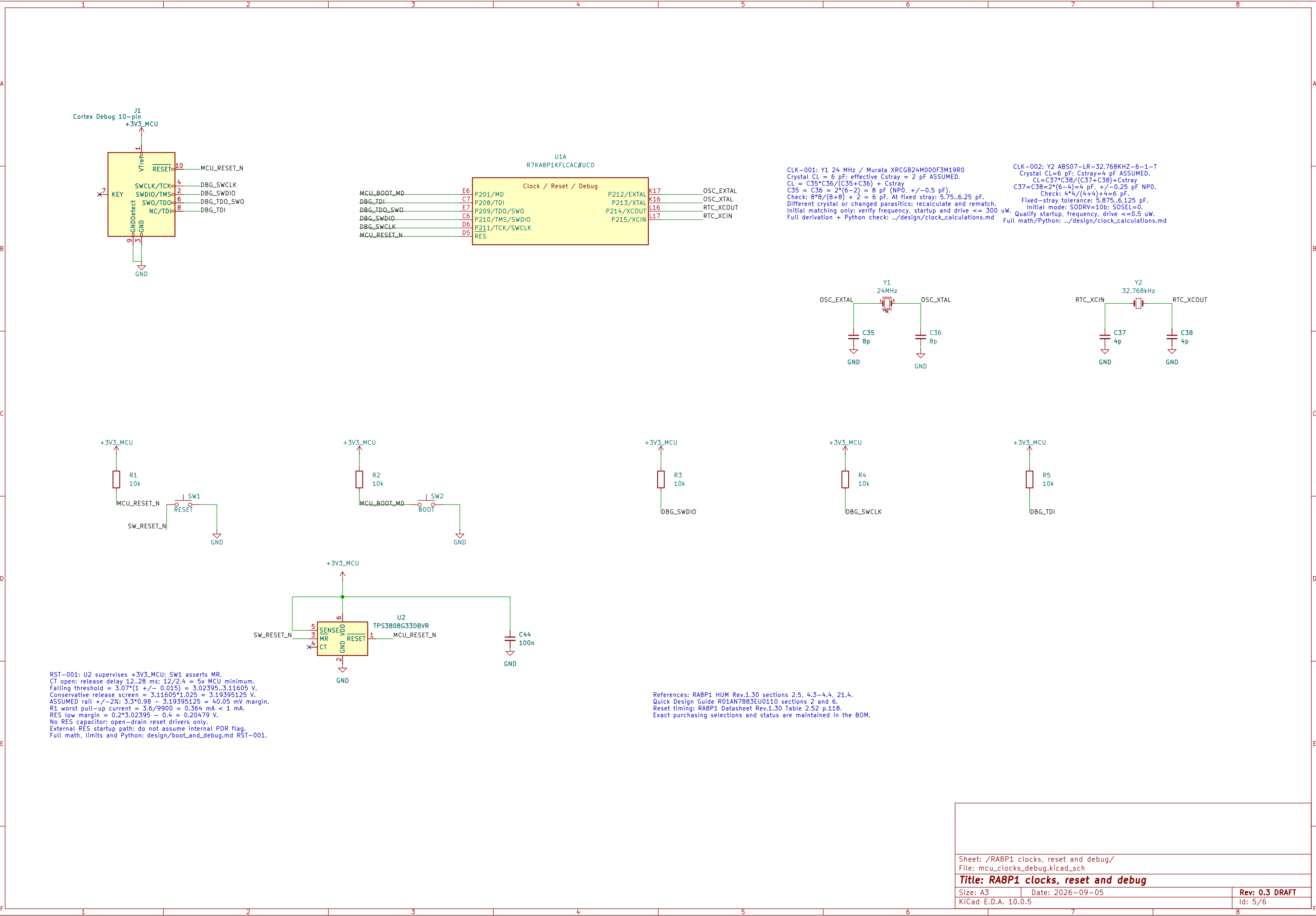
DC/DC

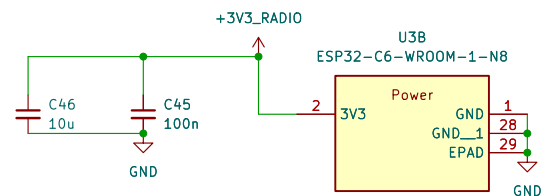
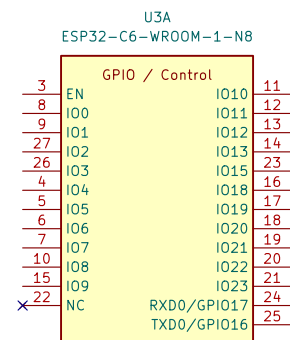
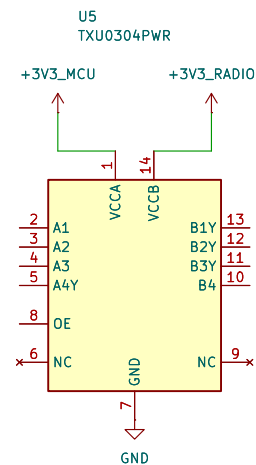
A10 VCC_DCDC VLO A7
A11 VCC_DCDC VLO A8
B10 VCC_DCDC VLO B7
B11 VCC_DCDC VLO B8
A9 VCC_DCDC
B9 VSS_DCDC
VSS_DCDC

GND

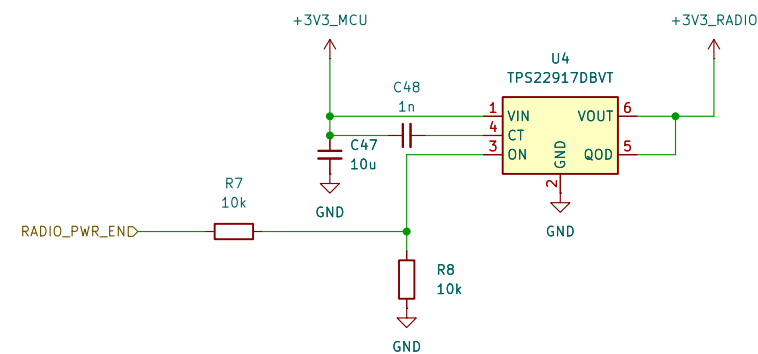








RADIO-003; C46 bulk bypass / TDK C16X7R1V106K160AC
 10 uF nominal, $\pm 10\%$, X7R; DC bias reduces capacitance,
 $C(3.3\text{ V}) = 9.69875 + (3.3 - 3.15) \cdot 0.85 \cdot (9.38950 - 9.69875)$
 $= 9.64448\text{ uF}$, $C(3.6\text{ V}) = 9.53503\text{ uF}$ (nominal).
 Source curve interpolation, NOT a guaranteed minimum.
 Rail: 3.0..3.6 V; source capability $\geq 0.5\text{ A}$.
 Supply and transmit transient qualification remain required.
 Calculations: ..\design\radio\interface.md#radio-003-bulk-capacitance



RADIO=004: U4 TPS22917DBVT / radio power gating
QOD tied to VOUT: Rdis = 150 ohm TYPICAL.
C45+C46 = 0.1+0.1 = 10.1 uF nominal external load.
t90...10 = ln(9)*150*10.1u = 3.329 ms (ideal RC).
Not a guaranteed off-time: module C/load and VIN decay matter.
C48 connects CT to VIN, not GND: 50 V rating >= 7 V required.
Host control, reset/isolation and upstream regulator remain incomplete.
Math + limits: ../design/radio_interface.md#radio-004-radio-load-switch

RADIO-005: C47 / U4 VIN local bypass
 10 uF nominal, +/-10%, 3.5 V X7R; same exact TDK part as C46.
 $C(3.3\text{ V}) = 9.69875 + (3.3 - 3.5) / 0.85 / (9.38950 - 9.69875)$
 $= 9.64418$ uF. Not a guaranteed minimum.
 Droop = $\text{integral}(\text{load} - \text{source}) / \text{dt} / \text{Ceff} + \text{ESR} / \text{ESL effects}$.
 TL section 11: 1 uF usually sufficient; slow sources may need more.
 C47 is upstream: exclude it from C45+C46 output-discharge math.
 Regulator/transient qualification required; no source flag added.
 Calculators: `./design/radio_interface.mdf` RADIO-005-input-bypass

```

RADIO-006: R7/R8 / active-high enable, default OFF
R7 series = R8 pulldown = 10k 1%; +/-2% screening envelope.
TP25291: ON; Voh = 0.35 V, Vih = 1.0 V
Assumed node leakage <= 10 uA and host VOL <= 2.5 V, VOH>=2.5 V.
Rparallel <= 5.1k; divider fraction = 0.49, 0.51.
Vlow <= 0.5*0.51+10u*5.1k = 0.306 V < 0.35 V.
Vhigh >= 2.5*0.49-10u*5.1k = 1.174 V > 1.0 V.
Hi-Z screen: 100u/10.2k = 0.102 V; disable host pull-up.
Host pin, all power states, leakage and timing need qualification.
Full math: ../designer/radio/interface.md#radio-006-enable-divisor

```

RADIO-2007: C48 = CT slow-rate control, TDK C1608N01H102/080AA
1nF +/-5%, 50 V NPO; CT-to-VIN, NOT CT-to-GND.
TI 3.6 V / 25 C TYPICAL table; not test, guaranteed at 3.3 V:
Slew = 1000/1000 = 1.9 mV/us; $\tau = 3.8 \times 1000 = 3.8$ ms;
IR(30, 90%) = 1.6 $\times$ 1000 = 1.6 ms (TI test)
C45+C46 = 10.1 uF nominal; Icap = C*dV/dt = 19.19 mA
Open-CT comparison: 10.1u/44mV/us = 444.4 mA (typical).
Total Inrush also includes module C/load and future rail parts.
Not a problem for reset, but for reset delay, capacitor regulation required.
Full math: $\therefore \text{design/radio_interface_mdf/radio-2007-slew-rate-capacitor}$

Incomplete electrical design – not for manufacture

Sheet: /ESP32-C6 radio/
File: radio_esp32.kicad_sch

Title: ESP32-C6 radio

Size: A3

Date: 2026-09-05

Rev: DRAFT

KiCad E.D.A. 10.0.5

Id: 6/6